

Full Stack Web and Mobile Web Development

Bootcamp Syllabus

Learn front-end and hybrid mobile development, with server-side support, for implementing a multi-platform solution.

The first two courses in this Bootcamp cover front-end frameworks: **Bootstrap** and **React**. You'll also learn to create hybrid mobile applications, using the **React Native**. On the server side, you'll learn to implement NoSQL databases using **MongoDB**, work within a **Node.js** environment and **Express** framework, and communicate to the client side through a RESTful API. Learners enrolling in this Bootcamp are expected to have prior working knowledge of HTML, CSS and JavaScript.

4 Courses	Project Portfolio	Certificate
At a structured pace, with the support of a dedicated Instructor and classroom environment.	Develop a restaurant website throughout the bootcamp, and work on your own project in groups or 2-3 students.	Showcase your accomplishment after each course and the end of the Bootcamp.

Front-End Web UI Frameworks and Tools: Bootstrap

Commitment

5 weeks of study, 8-10 hours/week

About the Course

This course will give you an overview of client-side web UI frameworks, in particular Bootstrap 4. You will learn about grids and responsive design, Bootstrap CSS and JavaScript components. You will learn about CSS preprocessors, Less and Sass. You will also learn the basics of Node.js and NPM and task runners like Grunt and Gulp.

At the end of this course, you will be able to

- a) Set up, design and style a web page using Bootstrap 4 and its components,
- b) Create a responsive web page design, and
- c) Make use of web tools to setup and manage web sites.

WEEK 1

Front-end Web UI Frameworks Overview: Bootstrap

This module gives you a quick introduction to full-stack web development and the outline of the course. Then you will learn the basics of Bootstrap, setting up a web project using Bootstrap. You will learn about responsive design and the Bootstrap grid system. At the end of this module, you need to complete your first assignment.

WEEK 2

Bootstrap CSS Components

This module concentrates on Bootstrap's components that are designed using pure CSS classes. You will learn about Navigation and Navigation bar. Then, you will learn about buttons, forms, tables, cards, images and media, tags, alerts and progress bars. At the end of this module you need to complete your second assignment.

WEEK 3

Bootstrap JavaScript Components + JavaScript Review 1

This module concentrates on Bootstrap's JavaScript based components. You will learn about tabs, pills and tabbed navigation, collapse, accordion, scrollspy, affix, tooltips, popovers, modals and the carousel. At the end of this module you need to complete the third assignment.

WEEK 4

Web Development Tools + JavaScript Review 1

This module rounds out our discussions on Bootstrap JavaScript components. Then we discuss CSS preprocessors, Less and Sass. Finally, we discuss building and deployment of our Web



projects through task automation using NPM scripts, and task runners like Grunt and Gulp. The final assignment of this course needs to be completed at the end of this module.

WEEK 5

Portfolio Project + JavaScript Advanced

This week, you will aim to complete your Bootstrap Portfolio Project, or as much of it as possible. You will also write a final report about your Portfolio Project, and you will share both your project and the report during the end-of-week workshop. Alongside this, you will begin to learn more advanced concepts in JavaScript, to help you prepare for the next course of your bootcamp, React.

Front-End JavaScript Frameworks: React

Commitment

5 weeks of study, 14-20 hours/week

About the Course

This course explores Javascript based front-end application development, and in particular the React library. This course will use JavaScript ES6 for developing React application. You will also get an introduction to the use of Reactstrap for Bootstrap 4-based responsive UI design. You will be introduced to various aspects of React components. You will learn about React router and its use in developing single-page applications. You will also learn about designing controlled forms. You will be introduced to the Flux architecture and Redux. You will explore various aspects of Redux and use it to develop React-Redux powered applications. You will then learn to use Fetch for client-server communication and the use of REST API on the server side. A quick tour through React animation support and testing rounds off the course. You must have preferably completed the previous course in the specialization on Bootstrap 4, or have a working knowledge of Bootstrap 4 to be able to navigate this course. Also a good working knowledge of JavaScript, especially ES 5 is strongly recommended.

At the end of this course you will:

- a) Be familiar with client-side Javascript application development and the React library
- b) Be able to implement single page applications in React
- c) Be able to use various React features including components and forms
- d) Be able to implement a functional front-end web application using React
- e) Be able to use Reactstrap for designing responsive React applications
- f) Be able to use Redux to design the architecture for a React-Redux application

WEEK 1

Introduction to JavaScript ES 6

In this module you are introduced to the latest JavaScript evolution and how Typescript supplements the language with string typing and additional debugging capabilities.

WEEK 2

Introduction to React

In this module we get a quick introduction to front-end JavaScript frameworks and libraries, followed by an introduction to React. We will also learn about React components and JSX.

WEEK 3

React Router and Single Page Applications

In this week, you will learn about various component types. You will learn about React router and its use in designing single page applications. You will also learn about single page applications and use React Router to design single page applications.

WEEK 4

React Forms, Flow Architecture and Introduction to Redux

In this module you will be introduced to uncontrolled and controlled forms and briefly examine form validation in React applications. You will get an overview of the Flux architecture and introduced to Redux as a way of realizing the flux architecture.

WEEK 5

More Redux and Client-Server Communication

In this module you will explore Redux further including Redux action, combining reducers, and Redux thunk, client-server communication using Fetch and the REST API. You will get a brief introduction to animation in React. You will also learn about testing, building and deploying React applications.

Multiplatform Mobile App Development with React Native

Commitment

4 weeks of study, 14-20 hours/week

About the Course

This course focuses on developing truly cross-platform, native iOS and Android apps using React Native and the Expo SDK. React Native uses modern JavaScript to get truly native UI and performance while sharing skills and code with the web. You will learn about UI development with React Native UI and layout support and access the native mobile platform's capabilities from Javascript using the Expo SDK. You should have already completed the Bootstrap 4 and the React courses in this specialization before proceeding with this course.

At the end of this course you will be able to:

- a) Build mobile applications targeting multiple platforms with a single codebase,
 - b) Leverage your HTML5, CSS, JavaScript and React skills, and
 - c) Use various features of the React Native and Expo SDK to build hybrid mobile applications
 - d) Use Redux to design the architecture for a React-Redux application
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WEEK 1

Hybrid Mobile App Development Frameworks: An Introduction

This module introduces you to hybrid mobile application development. You will learn about React Native and explore some of the features of React Native to implement a mobile app based on the React application that was implemented in the previous course on React.

WEEK 2

React Native UI Elements and Redux

This module introduces you to various React Native UI elements. We will look at how we can make use of these elements in designing the various views of our application. You will get an overview of the Flux architecture and introduced to Redux as a way of realizing the Flux architecture.

WEEK 3

React Native Alerts, Animations, Gestures, and Persist Redux Store

In this module we look at enhancing the user experience through the use of animations and support for gesture-based interaction. We also look at persisting the redux state, and alerting the users.

WEEK 4

Accessing Native Capabilities of Devices: The Expo SDK

In this module you will explore the Expo SDK that enables you to access the native capabilities of the mobile devices. You will use a few SDK APIs in order to understand the general concepts and the patterns for using these APIs within your React Native application.

Server-side Development with NodeJS, Express and MongoDB

Commitment

4 weeks of study, 14-20 hours/week

About the Course

This course deals with all things server-side. We base the entire course around the NodeJS platform. We start with a brief overview of the Web protocols: HTTP and HTTPS. We examine NodeJS and NodeJS modules: Express for building web servers. On the database side, we review basic CRUD operations, NoSQL databases, in particular MongoDB and Mongoose for accessing MongoDB from NodeJS. We examine the REST concepts and building a RESTful API. We touch upon authentication and security. Finally, we review backend as a service (BaaS) approaches, including mobile BaaS, both open-source and commercial BaaS services.

At the end of this course, you will be able to:

- a) Demonstrate an understanding of server-side concepts, CRUD and REST
 - b) Build and configure a backend server using NodeJS framework
 - c) Build a RESTful API for the front-end to access backend services
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WEEK 1

Introduction to Server-side Development

In this module you will be introduced to Node, Node modules and the Node HTTP server. You will learn about the Express framework and how to set up a REST API using Express.

WEEK 2

Data, Data, Where art Thou Data?

This module looks in detail at data storage with MongoDB, the popular NoSQL database. You will learn first about Express generator for scaffolding an Express application. Then you will learn about MongoDB. You will learn how to interact with MongoDB from a Node application. Then you will learn the Mongoose ODM to create schemas and models, and interact with MongoDB server.

WEEK 3

Halt! Who goes there?

This module is dedicated to user authentication. We first develop a full-fledged REST API server with Express, Mongo and Mongoose. Thereafter we examine basic authentication and session-based authentication briefly. We then develop token-based authentication with the support of JSON web tokens and the Passport module.

WEEK 4

Backend as a Service (BaaS)

In this module we learn about Mongoose population, a way of cross-referencing documents and populating the documents from other documents. We then review secure communication using HTTPS. We look at Backend as a Service (BaaS) and take a brief look at Loopback.

Project Portfolio

Commitment

Self-driven and in groups, 4-8 hours/week for the duration of the Bootcamp

About the Project Portfolio

This Bootcamp offers a structured path for you to develop a fully functional and feature rich project portfolio. You will be asked to start with identifying the project you would like to build, based on a market analysis of needs, and will quickly move to a prototyping phase. With every new course and technology, you will improve your project to make it more responsive, data-driven, mobile-friendly and finally connect it with a back-end and host it on a cloud platform.

You will be invited to work in groups and offered a weekly rhythm to progress.